Machine-Learning Wireless Intrusion Detection System

Detecting 802.11 Deauthentication Floods with a Random Forest

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A Random Forest looks at each MAC address every second and flags deauthentication floods. On held-out traffic it got 100% accuracy and named the MAC that caused the flood.

The problem

Deauthentication frames are management frames, and many networks do not protect them. An attacker can spoof them and kick clients off the network. That is a denial of service, and it can also help an evil twin attack. A **WIDS** detects this and raises an alert. A WIPS would also block it. I built the WIDS.

Dataset

I used one capture from the public **WPA3 Attacks Dataset** (Mendeley Data, CC BY 4.0): the deauthentication attack capture.

- 916 s, 722,606 frames
- 515,384 normal frames
- 207,222 deauth-labelled frames
- 370 s flood; peak over 1,300 deauth/s
- Isolated WPA3 testbed built by the dataset authors
- No new traffic captured or generated

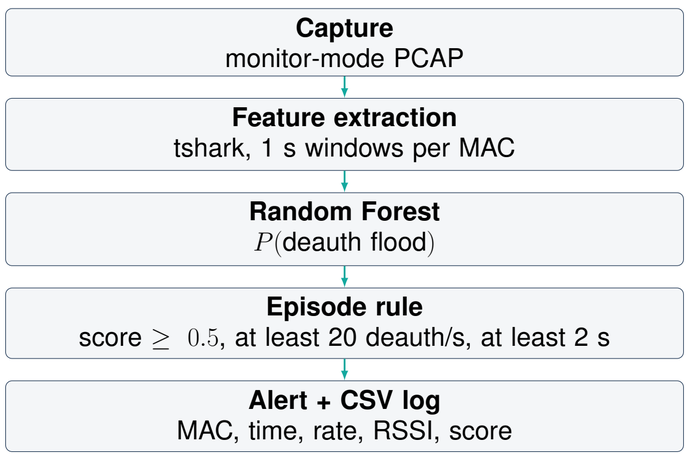
Citation: Halbouni, Ong & Leow (2026), *WPA3 Attacks Dataset*, DOI 10.17632/cxx5t5nw7z.3, CC BY 4.0.

Features

I made one row per **transmitter MAC per second**: 2,692 rows, 235 MACs and 22 features. The model does not get the MAC address or the timestamp.

Feature	Why it matters
deauth count / ratio	Direct flood signal
frames per second	Flood volume
frame-type counts	Management vs data mix
mean / std RSSI	Transmitter proximity
unique destinations	Broadcast / spoof pattern
retry ratio	Channel stress
reason-code count	Deauth/disassoc variety

System architecture



Machine learning

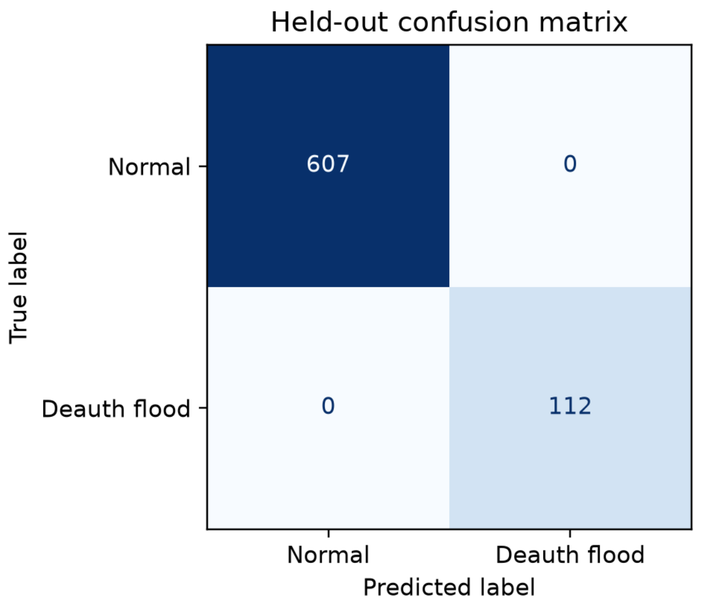
- Random Forest: 300 trees, depth 10, balanced class weights.
- Baseline: Decision Tree, depth 6.
- Median imputation for missing RSSI.
- **Time-based split:** earlier traffic for training, later traffic held out. No random window leakage.
- 1,973 training windows and 719 test windows (607 normal, 112 attack).

Results

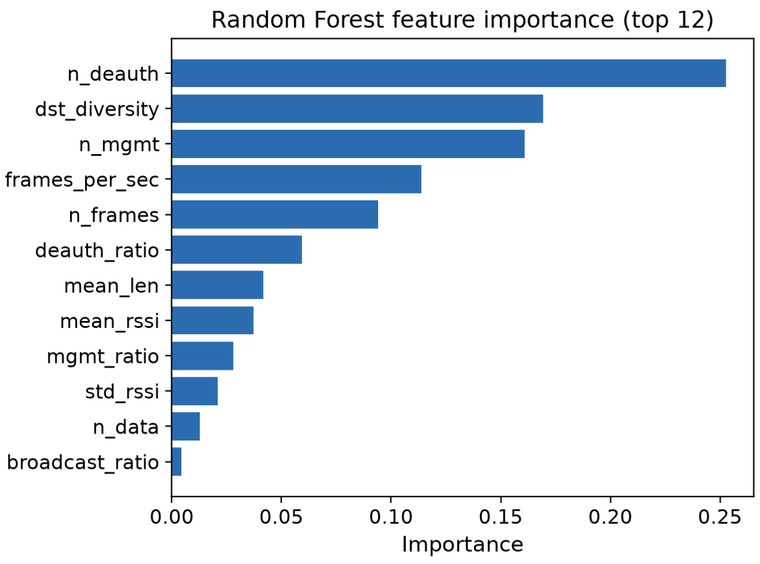
Metric	Random Forest	Decision Tree
Accuracy	100.0%	100.0%
Precision	100.0%	100.0%
Recall	100.0%	100.0%
F1	100.0%	100.0%

Confusion matrix (RF): TN 607, FP 0, FN 0, TP 112.

The flood is easy to separate. Normal traffic still has 436 windows with some deauth frames (up to 64/s), but the attack sends hundreds to over 1,300/s. The lowest attack score was 0.85 and the highest normal score was 0.28.



Confusion matrix on held-out later traffic.



Random Forest feature importance (top 12).

Detection demo

60 s normal excerpt: **no alerts**.

60 s attack excerpt:

```
[ALERT 1] 07:34:12 to 07:35:07
MAC 02:00:00:00:00:72
deauth=215/s RSSI=-29 dBm score=1.00
[ALERT 2] 07:34:12 to 07:35:07
MAC 02:00:00:00:00:cb
deauth=1289/s RSSI=-32 dBm score=1.00
```

On the full 15-minute capture the detector found the two sustained floods. Two short low-rate exchanges (3 to 16/s) were filtered by the 20/s rule.

Limitations

- One capture, one testbed, so 100% is optimistic.
- 1-second windows, and only deauth floods are modelled.
- MAC spoofing makes the source in the alert unreliable.
- No live monitor-mode test because the VM has no adapter. The PCAP replay path is used.
- Two low-rate episodes are false positives at window level, but the rate rule suppresses them.

What I would improve

- Test on AWID3 and new captures from different networks.
- Add disassociation floods, beacon floods and evil twin detection.
- Try sliding windows and online learning for concept drift.
- Test evasion with low-rate, bursty or spoofed-source attacks.
- Add a live dashboard and maybe AP integration.

AI use

I used OpenCode with a DeepSeek model to help me find and download the public dataset, set up the Python pipeline, debug the tshark extraction and draft the poster text. I checked the dataset and code, ran the training and evaluation, and confirmed that no MAC address or timestamp is used as a feature. All reported numbers come from the included scripts.

Reproduce

```
python prepare_data.py
python train.py
python detect.py --pcap data/demo_attack.pcap
```

Code and results: local submission.

Dataset DOI: 10.17632/cxx5t5nw7z.3. Python, tshark, scikit-learn.

Bottom line

A Random Forest on per-MAC, per-second features detects deauthentication floods with 100% accuracy on held-out traffic and gives a MAC-level alert.

Key numbers

722,606 frames
207,222 deauth
2,692 windows
100% accuracy